

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Any two (x1) from - Many {alveoli/capillaries} / folded alveoli + {provide a large surface area/ increasing <u>rate</u> of diffusion} (1) {Squamous epithelium/alveolar walls/ capillary walls} are {thin/ one cell thick} + {short diffusion pathway/ increasing <u>rate</u> of diffusion} (1) Reject thin cell walls {Many capillaries/ good blood supply/ capillary network} + {maintain (steep) concentration gradients/ increasing <u>rate</u> of diffusion} (1)	2			2		2
		(ii)		(Breakdown of the alveolar walls) reduces the surface area (of the alveoli) (1) - Less oxygen <u>diffuses</u> into the blood/ decreases the rate of <u>diffusion</u> (of oxygen) into the blood (1)		1	1	2		2
	(b)	(i)		Any four (x1) from A. (External) Intercostal muscles contract and the <u>rib</u> cage moves {upwards / outwards / expands} (1) B. The diaphragm contracts and {flattens/ goes down} (1) C. Pulls on pleural membranes (1) D. Increased {thoracic/ thorax/ lung/ alveolar} volume (1) E. Which decreases {lung/alveolar} pressure (1)	4			4		

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						AO1	AO2	AO3	Total	Maths	Prac
		(ii)		<u>Difference</u>	<u>Explanation</u>						
				Forced expiration is slower in {emphysema/4.4}/ ORA (1)	narrowing of bronchioles/ lumen decreases (due to mucus/ thickening of wall) (1)						
				{increased residual volume / forced vital capacity smaller/ less air is forced out of the lungs} in {emphysema/4.4}/ ORA (1)	elastic <u>recoil</u> is lost / elastic <u>recoil</u> is less effective (1)			4	4		2
				Question 4 total		6	1	5	12	0	6